XAUUSD
PULSE

Project Documentation

Requirements | Design | Implementation | Testing | Deployment | Maintenance

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Analysis tool for personal and educational use. Not financial advice. No promise of profit.

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1. Executive summary

1.1 What it is

XAUUSD Pulse is a personal, live gold (XAU/USD) intelligence dashboard. It shows a real-time price, explains how each news headline is likely to affect gold and the US dollar, generates rule-based trading signals from four strategies, and, most importantly, tests those strategies honestly before anyone trusts them. It runs on a normal Windows PC for $0, using free data sources and a local SQLite database.

1.2 Why it is different

Most dashboards stop at showing signals. This project is built around the harder question: **can the signals be trusted?** Every strategy goes through a seven-check Test lab (walk-forward testing on unseen data, a settings sweep, a higher-cost test and a bootstrap luck check) and is then followed live in a journal with nothing back-filled. A strategy is only reported as good to go (for demo-account testing) after it passes the lab on two runs at least 12 hours apart.

1.3 Status at a glance

Phase	Outcome	Status
1. Requirements	Goals, scope, constraints and success criteria agreed and recorded	Complete
2. Design	Architecture, data model, algorithms, interface and security design	Complete
3. Implementation	Working system in versions 1.0 to 2.0 (about 70 source and test files)	Complete
4. Testing	70 automated tests, integration checks, headless-browser checks, Test lab on real data	Ongoing (live journal needs weeks of data)
5. Deployment	Runs locally on Windows; GitHub package prepared	Local done; free hosting not yet built
6. Maintenance	Plan, checks and troubleshooting guide in this document	Plan ready

> **Honest headline result.** On the first real-data runs, the original strategy scored **Not proven** (2 of 7 checks passed). The lab showed that an apparently good result in raw points (profit factor 1.54 on H4) was largely an effect of gold's rising price level and of weekend bars; measured in risk units on cleaned data it was about -0.18R per trade. No strategy in this project is proven to be profitable. The system exists to find that out safely.

2. How the project was run

The project followed six phases, each agreed before the next began. Work was done as short iterations: build a piece, test it, show it, correct it. The table lists what each phase produced and the evidence for it.

Phase	What was produced	Evidence
Requirements	Purpose (analysis, news impact, signals, strategy performance), single user, free-only budget, live data, no chart; later additions: alerts, journal, calendar pause, calculator	Section 3; requirement table F0 to F15
Design	Node.js + Express server, SQLite, Server-Sent Events, rule-based news impact, R-multiple backtester, walk-forward lab, alert state machine, neon-glass interface	Section 4; architecture diagram; data model
Implementation	Versions 1.0 (live ticker, candles, signals, news) to 2.0 (new interface, documentation); four strategies, lab, journal, alerts, history loading	Section 5; version table; GitHub history
Testing	Unit and integration tests; fake servers for each data source; headless browser checks; lab runs on real data	Section 6
Deployment	START.bat for Windows, key entry in the page, GitHub-ready package, CI workflow	Section 7
Maintenance	Free-tier watch, backups, upgrade steps, troubleshooting table	Section 8

2.1 Working method

- **Small steps, always tested.** Each feature was exercised against local fake servers before it was handed over, because the development environment could not reach the real data providers.
- **Real feedback drove changes.** The owner's screenshots of the real system exposed three problems that tests alone had not: stale H4 results, weekend bars in the data, and a mislabelled comparison table. Each was fixed and covered by a new test.
- **Honesty over polish.** Where something is unproven (strategy edge, news impact, event pause) the interface and this document say so.
- **AI pair-programming.** The system was designed and built together with Claude (an AI assistant) for design, code review and testing; the owner directed scope, reviewed results and ran the system on real data.

3. Phase 1: Requirements

3.1 Goal

Give one person an always-current, honest view of what is moving gold: live price, the news behind it, signals, and evidence about whether those signals deserve trust.

3.2 Scope decisions

Question	Decision
Purpose	Market analysis, latest news affecting gold and the dollar and how, signals, strategy performance
Users	One user (personal tool). No accounts.
Budget	Free only: data, hosting (local first) and APIs
Data freshness	Live price required; candles near-live within free limits
Price chart	Not required, by choice. The interface uses numbers, gauges and a headline tape instead.
Presentation	Colourful, animated, high-visibility interface; published under the author's name

3.3 Functional requirements

ID	Requirement	Priority
F0	Live price ticker (price, change, high, low, live status) with automatic reconnect	Must
F1	Live and recent news on gold, USD, Fed, inflation, yields and geopolitics	Must
F2	Each headline tagged bullish, bearish or neutral for gold and for the dollar, with a short reason	Must
F3	Sentiment gauges for gold and the dollar from recent news	Must
F4	Signals (buy, sell, hold) with entry, stop loss and take profit	Must
F5	Strategy performance: trades, win rate, profit factor, drawdown, trade list	Must
F6	Economic calendar of upcoming high and medium impact US events	Should
F7	Four strategies selectable from one menu, all on one engine	Must
F8	Test lab: data quality, results by period, walk-forward, settings sweep, cost test, luck check, seven-check verdict	Must
F9	Live signal journal that records signals from now on and follows them to real outcomes	Must
F10	On-site alerts: good to go (two passes 12+ hours apart), live signals for passing strategies, warnings	Must
F11	Event pause around high-impact US events (live filter)	Should
F12	Position-size calculator	Should
F13	Up to 8 years of candle history loaded in the background	Should
F14	CSV export of backtest trades and the live journal	Could
F15	Dark and light themes, animation, mobile layout, reduced-motion support	Should

3.4 Non-functional requirements

ID	Requirement	How it is met
N1	Cost: $0 to run	Free data tiers, SQLite, local hosting
N2	Performance: page usable in about 3 seconds; live updates without refresh	Server-Sent Events; small REST payloads; lab runs in the background
N3	Resilience: a failing source must not break the site	Per-source error handling; last good data kept; status page
N4	Security and privacy: keys stay on the server; local access only	Section 4.7
N5	Honesty: unproven results must be labelled; fake data must never look real	Sample-data banners; sample results never raise alerts
N6	Maintainability: tested, modular code	70 tests; pure engine separated from I/O

3.5 Out of scope

- Trade execution or any broker connection
- Multiple users, accounts, payments
- A native mobile app
- Guaranteeing profit of any kind

4. Phase 2: Design

4.1 Architecture

A Node.js and Express server owns every connection to external data, stores results in SQLite and pushes live updates to the browser over a single Server-Sent Events stream. Centralising source access protects the free-tier limits, and keeps API keys away from the browser.

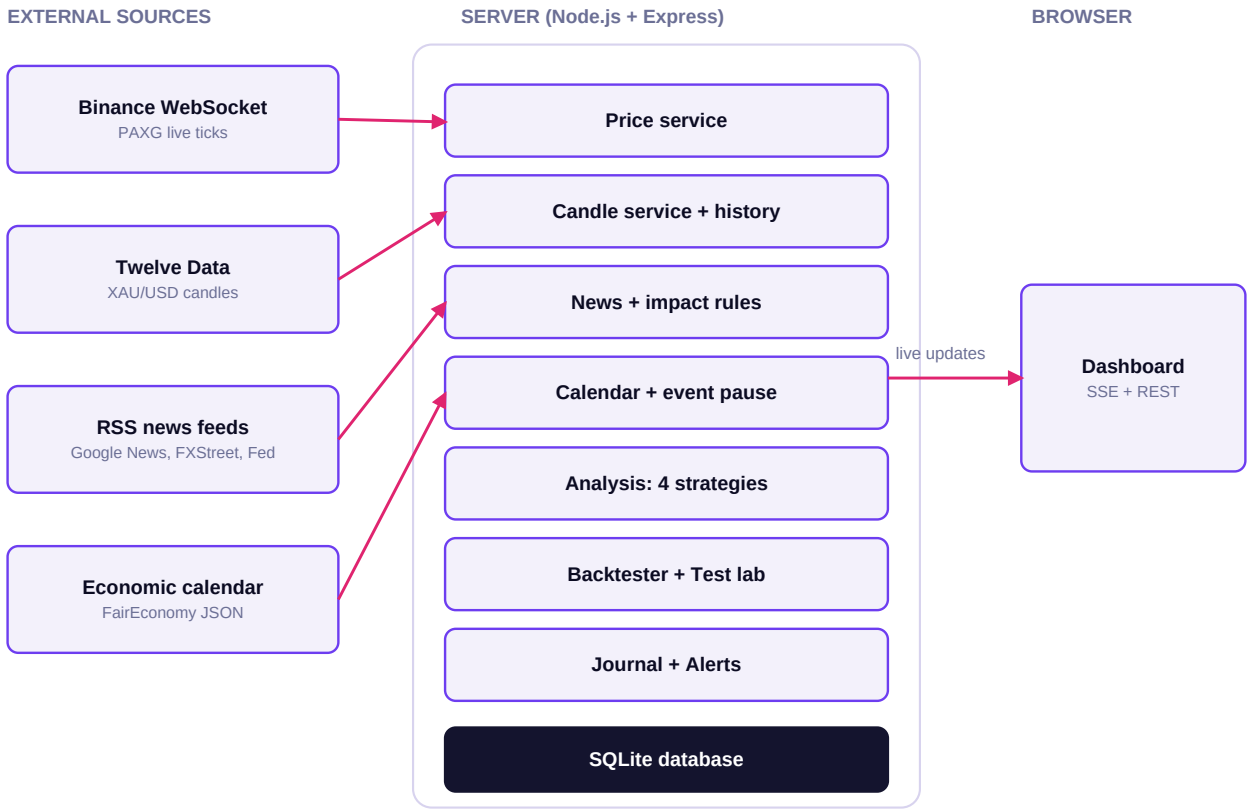


Figure 1: System architecture

4.2 Technology choices

Layer	Choice	Reason
Server	Node.js 20+, Express	Good fit for live push and many small polling jobs
Database	SQLite (better-sqlite3)	Free, serverless, transactional; ample for one user
Live updates	Server-Sent Events	Simple one-way push; the browser reconnects by itself
Interface	Vanilla JavaScript, CSS animations	No build step; fast; easy to read and maintain
Fonts	Space Grotesk, Inter, JetBrains Mono (bundled)	Same look offline; no external requests
Testing	node:test, headless Chrome, local fake servers	No extra frameworks; realistic end-to-end checks

4.3 Data sources and the free-tier budget

Data	Source	Notes
Live price	Binance public WebSocket, PAXG/USDT	No key. PAXG is a gold-backed token: close to spot gold but not identical
Candles	Twelve Data free plan (XAU/USD)	Key needed. About 220 requests a day (limit 800). Loading 8 years of history is a one-off of roughly 15 requests, paced at 8.5 seconds each
News	Google News RSS searches, FXStreet, Federal Reserve	Polled every 2 minutes; keyword-filtered; duplicates removed by URL and title
Calendar	FairEconomy weekly JSON	US high and medium impact events; refreshed twice a day

Free-tier terms change. Every limit above must be re-verified before relying on it.

4.4 Data model

Table	Holds
candles	OHLC bars per timeframe (primary key: timeframe + bar time)
news, news_impact	Headlines (unique URL) and their gold and dollar bias, score and reason
signals	Signals produced by each strategy
backtest_runs, backtest_trades	Latest backtest per strategy and timeframe, with every trade
journal	Live signals and outcomes: status, entry, exit, R, event-pause flag (unique per strategy, timeframe, signal time)
notifications	On-site alerts with level, kind, read flag and a de-duplication key
events	Economic calendar (unique per time, name, currency)
meta	Small facts: data source in use, journal start times, stored lab results, history-loaded flags

4.5 Key algorithms and rules

Signals and the backtest engine

- A signal is decided at a bar's close. The trade enters at the **next** bar's open, so the engine cannot peek at the future.
- Stop loss and take profit come from ATR(14): stop = multiple x ATR; target = reward ratio x stop.
- One trade at a time. If one bar touches both stop and target, the stop is assumed to hit first (the pessimistic choice).
- Every trade pays a cost (0.35 points by default). Results are reported in points and in **R**, where 1R is the amount risked, so different years compare fairly.
- A test proves each strategy's signal at a bar does not change when later bars are added (no look-ahead).

The four strategies

Strategy	Rule in plain words
Trend pullback v1	Fast average above slow (21/50) and RSI turns back up through 40: buy; mirror for sell
Trend pullback v2	Slower trend (21/100), RSI 45/55, 2x ATR stop, 3R target. Chosen after seeing history, so lab results are optimistic
Breakout	Close above the highest high of the last 20 bars while above the 100-bar average: buy; mirror for sell
Oversold bounce	RSI turns back up through 30 while price is below its 20-bar average: buy; mirror at 70

The Test lab and its verdict

The lab removes weekend and flat bars, then asks seven questions. Each is a plain PASS or FAIL:

#	Check	Pass rule
1	Profitable after costs on all data	Average R above 0 and profit factor above 1
2	Unlikely to be luck	90% or more of 2,000 resampled histories are profitable
3	Survives double costs	Profit factor above 1 at 0.7 points per trade
4	Steady over time	60% or more of quarters (with 3+ trades) are profitable
5	Works across nearby settings	60% or more of up to 243 settings combinations are profitable
6	Profitable on unseen data	Walk-forward over three unseen periods: 20+ trades and average R above 0
7	Large enough sample	100 or more trades

Verdict: **Promising** (6 to 7 passes), **Mixed** (4 to 5), **Not proven** (0 to 3). Walk-forward tunes settings on the past with a t-statistic objective, then tests them on the next unseen block; the original settings are tested on the same unseen blocks for comparison.

Alerts, good to go, and the journal

- **Good to go** needs a Promising verdict on two lab runs at least 12 hours apart, on real data. It means ready for a demo account.
- **Live signal alerts** are sent only for strategies that currently pass; signals inside an event-pause window are sent as on hold.
- **The journal** records signals from the moment it starts (no back-filling), applies the same one-trade-at-a-time rule as the backtest, and after 20 completed trades posts a live-results message.

- **Sample data safety:** results computed on sample data never raise alerts, and are deleted when real data replaces them.

News impact and sentiment

Headlines are scored by transparent regular-expression rules (for example: dollar rises, yields climb, rate cuts, safe-haven demand). Each rule has a gold weight, a dollar weight and a plain-English reason. Sentiment is a recency-weighted average of the last 24 hours, shrunk toward neutral when few headlines have a clear driver. The rules read headline text only and can misjudge ambiguous headlines.

4.6 Interface design

The interface was redesigned for a younger, attention-grabbing look while keeping the numbers readable. It uses a dark neon-glass theme by default with a light theme one tap away.

- **Colour and type:** gold, pink and violet accents on deep navy; green and pink for bullish and bearish; headings in Space Grotesk, body in Inter, numbers in JetBrains Mono.
- **Motion that signals life:** animated gradient background, glowing price that flashes on every move, ripple on the live badge, a scrolling headline tape, sliding-in news, gauges that glide, pulsing BUY and SELL badges, a ringing alert bell and confetti on good to go.
- **Accessibility:** high contrast text, labels for controls, and a prefers-reduced-motion setting that switches animations off.
- **Responsive:** one column on phones; the header actions wrap; no horizontal scrolling.

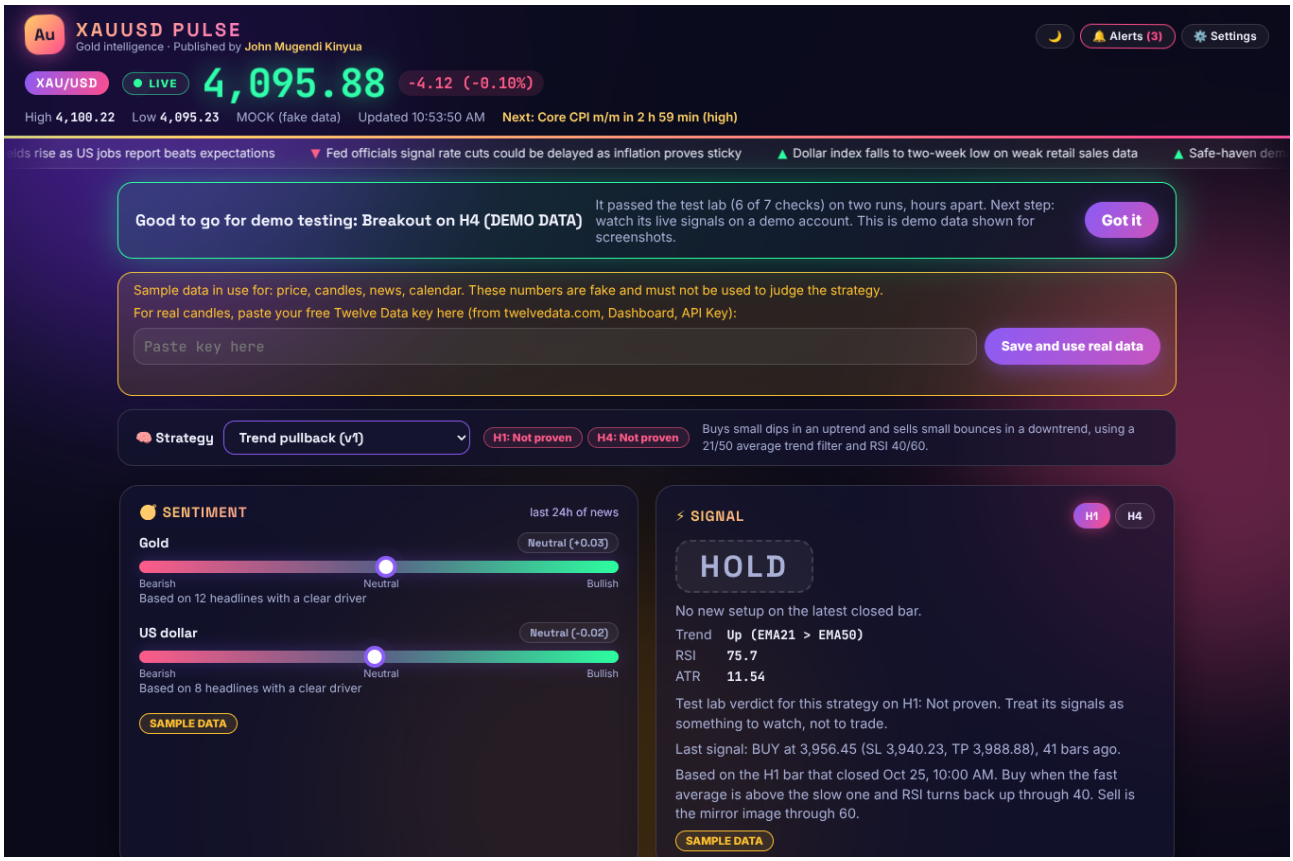


Figure 2: Dashboard, dark theme (demo data)

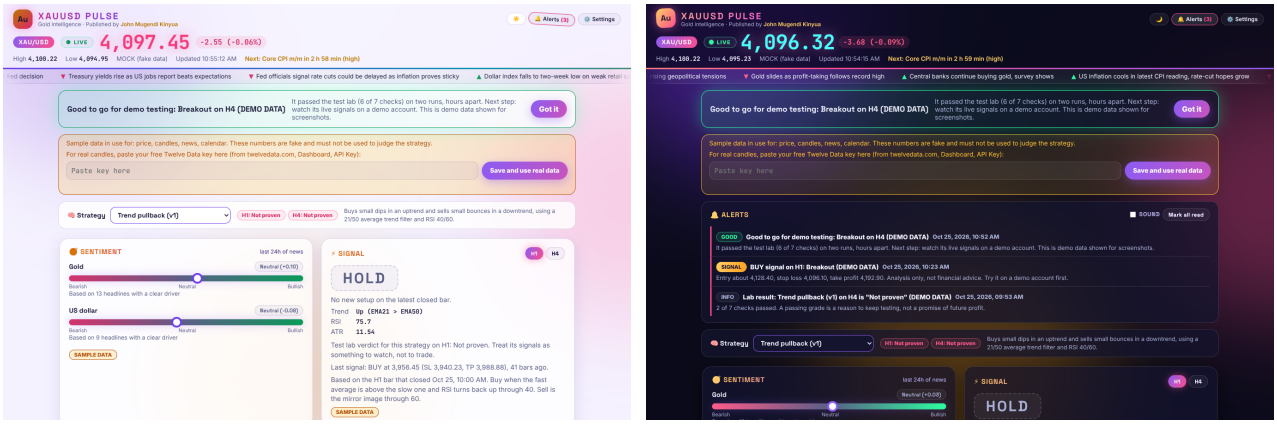


Figure 3: Light theme (left) and the alerts panel (right). Demo data.

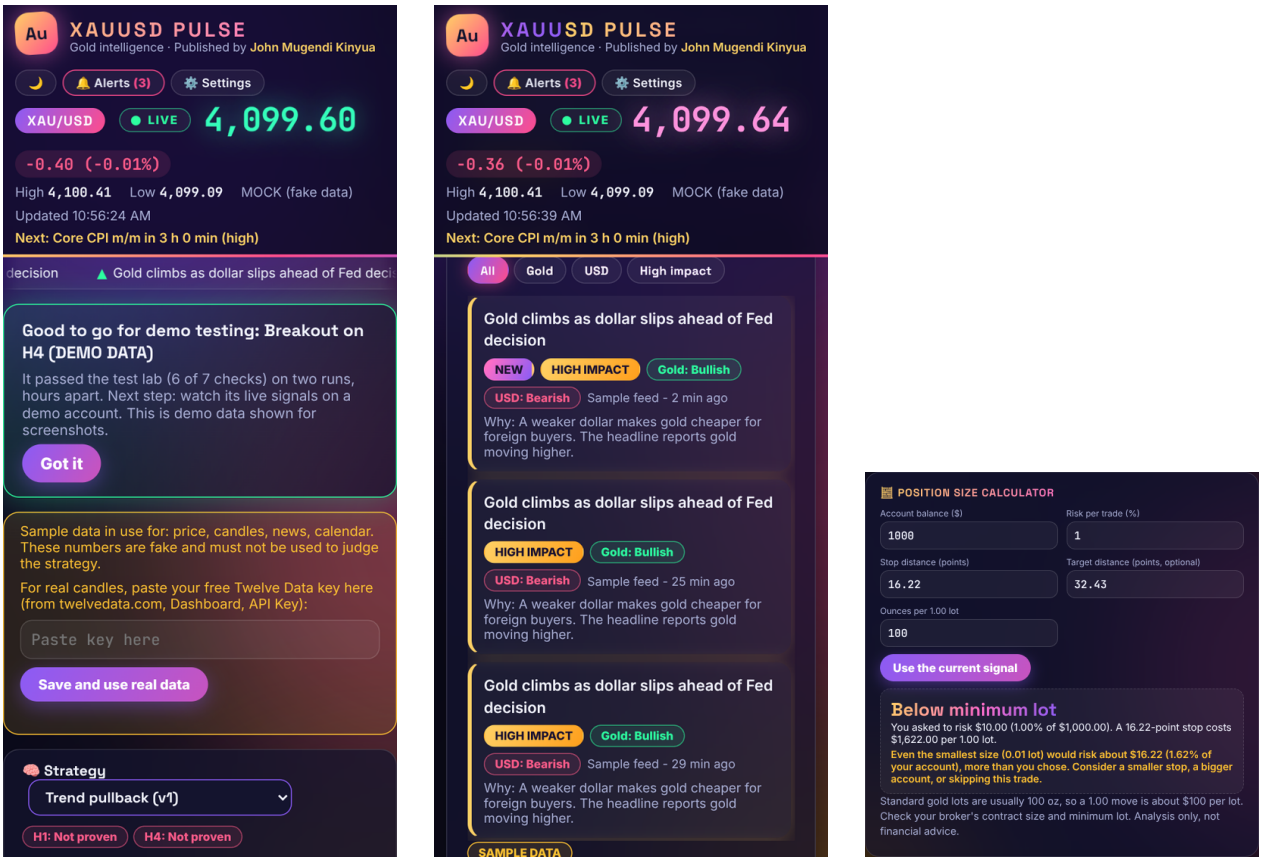


Figure 4: Phone layout (two views) and the position-size calculator. Demo data.

4.7 Security and privacy design

- The server listens on 127.0.0.1 only; no accounts, tracking or cloud services.
- The Twelve Data key stays on the server and in a git-ignored .env file; it is never sent to the browser or logged, and error messages have it masked.
- Endpoints that change state (saving the key, starting the lab, marking alerts read) reject non-local Host headers and foreign Origins, blocking DNS-rebinding and cross-site requests.
- Text from news feeds is inserted into the page with textContent, never as HTML, and links are limited to http and https.
- Before any public hosting, add authentication, HTTPS and rate limiting (see section 7).

5. Phase 3: Implementation

5.1 Version history

Version	What was added
1.0	Live Binance ticker, Twelve Data candles, RSI + EMA signal engine, backtester, news with impact tags, sentiment, calendar
1.1	Key entry inside the page; sample and real candles are never mixed
1.2	Data check page, retry after failed backtests, 5,000-bar history, Settings panel
1.3	Test lab: walk-forward, settings sweep, cost test, luck check, verdict
1.4	Four strategies, on-site alerts, good-to-go rule, live journal
1.5	8-year history loading, event pause, position-size calculator, next-event countdown, sound, CSV export
2.0	New neon-glass interface with publisher credit; README, GitHub package, CI, this document

5.2 Module guide

Area	Files	Responsibility
Server	server.js, api.js, config.js, db.js, bus.js	Wiring, SSE stream, REST, settings, schema and upgrades
Data	services/price.js, candles.js, news.js, calendar.js	Each external source, with retries, pacing and error handling
Analysis	services/analysis.js, backtest.js, journal.js	Signals, backtests, live journal
Strategy	strategy/*.js, strategy/engine.js	Four strategies; pure backtest engine with R-multiples
Lab	lab/lab.js, stats.js, dataQuality.js, status.js	Statistics, walk-forward, verdict and good-to-go logic
Alerts	services/notifications.js, labService.js, eventPause.js	On-site messages, scheduled lab runs, pause windows
Interface	public/index.html, styles.css, app.js, calc.js	Dashboard, animation, position calculator

5.3 Defects found and fixed during the project

Recording these is part of the engineering story: each was found by testing or by the owner's real-data screenshots, fixed, and covered by a regression test.

Defect	Effect	Fix
Sample and real candles could mix after adding a key	Fake weekend bars would corrupt real signals	Stored data source is tracked; everything derived is purged when it changes
Stale backtest after a failure	H4 showed 0 trades although signals existed	Result is remembered only after success, so failures retry; failures are visible in status
Sample results could raise good-to-go alerts	Fake data looked like real success	Sample results are excluded from alerts and discarded on switch
Strategy name and verdict shared a field name	Comparison table showed verdicts as strategy names	Fields renamed; regression test added
Progress bar could move backwards	Confusing lab progress	Progress is monotonic; test added

Defect	Effect	Fix
Raw points flattered recent years	Profit factor 1.54 hid a break-even result in R	Lab reports in R, cleans weekend bars, runs by period

6. Phase 4: Testing

6.1 Approach

Because the live providers could not be reached from the development environment, each source has a sample mode and a local fake server that mimics its real format, including failures (rate limits, broken feeds, rejected keys). On top of unit tests, the whole system was run end to end and the dashboard was checked in a headless browser.

6.2 Automated tests (70 passing)

Area	What is checked
Indicators	EMA, RSI, ATR, SMA and channel maths against known values and edge cases
Strategies	Well-formed signals (stop and target on the right sides), no look-ahead, consistent R, no overlapping trades, unique definitions
Backtest engine	Take-profit, stop-loss, time exit, stop-first rule, open trades at the end of data, metrics
News impact and sentiment	Headline cases for both directions, recency weighting, shrinkage
Test lab	Bootstrap repeatability, period grouping, data cleaning, grid validity, walk-forward never tests on training data, verdict adds up
Alerts	Candidate then confirmed then dropped transitions; de-duplication; sample data never alerts
Journal	No back-filling, no duplicates, outcomes identical to the backtest, skipped signals really overlap, event-pause flag
History and data	Paging backwards, stop rules, error pause, no repeat, sample anchoring
Other	Event pause windows, CSV escaping, position-size maths, key file editing

6.3 Integration and browser checks

- Server run against fake Twelve Data, RSS and calendar servers, including a rate-limit error, a broken feed and a wrong key; the key never appears in logs.
- Security checks: forged Host and foreign Origin requests are rejected with 403.
- A simulated browser and a real headless Chrome exercised strategy switching, alerts, the confetti banner, the calculator, theme toggle and exports with no JavaScript errors; fonts load; the phone layout does not scroll sideways.
- Performance: the lab takes about 8 seconds on 8 years of hourly data and never freezes the live price for more than about 100 ms.

6.4 Strategy validation on real data

The owner ran the Test lab on real Twelve Data candles on 23 October 2026 (H4, January 2024 to October 2026, weekend bars removed):

Measure	Result	Reading
Verdict	Not proven (2 of 7 checks)	No reliable edge for the original strategy
All-data result	34 trades, -0.18R per trade, profit factor 0.65	Slightly losing after costs
Luck check	20% of resampled histories profitable	Could easily be luck

Measure	Result	Reading
Steady over time	1 of 6 quarters profitable	Not steady
Costs doubled	Profit factor 0.64	Costs are not the problem; signal quality is
Settings sweep	64% of 231 settings profitable; original settings ranked 204th	The starting settings were among the worst
Walk-forward (tuned)	55 unseen trades, +0.17R per trade, 2 of 3 periods positive	The only encouraging sign; modest and unstable
Raw vs cleaned data	+0.03R on 44 trades vs -0.18R on 34 trades	Weekend bars changed the conclusion

What this teaches. A first look at H1 and H4 in raw points suggested an edge (profit factor 1.16 and 1.54). Longer data, risk units and cleaning removed it. That is exactly the failure the lab was built to catch. The follow-up work (more history, version 2 settings, the live journal) is the proper way to test improved ideas without fooling ourselves.



Figure 5: Test lab (demo data)

6.5 Known gaps

- No strategy has yet a track record in the live journal; that takes weeks of running.
- Event pause cannot be backtested because the calendar has no history; the journal tags such trades so the rule can be judged later.
- News impact is rule-based on headlines only and has not been measured against price reaction.
- Real-provider behaviour (rate limits, outages) was tested with fakes, and then in the owner's live runs for price, candles, news and calendar.

7. Phase 5: Deployment

7.1 Running locally (Windows)

- Install Node.js LTS from nodejs.org.
- Extract the zip, then double-click START.bat. The first run installs dependencies (about a minute) and opens http://127.0.0.1:3000.
- Click Settings and paste the free Twelve Data key, or run SET-KEY.bat. History loads in the background.
- Keep the black window open while using the site; closing it stops the platform. START.bat also stops an older copy still running on port 3000.

7.2 Publishing to GitHub

The repository package contains the README, licence (MIT), changelog, CI workflow (runs the tests on Node 20 and 22), screenshots, this document and a step-by-step guide (docs/PUBLISH_TO_GITHUB.md). The .env file and the data folder are git-ignored, so the API key is never published.

7.3 Configuration reference

Variable	Default	Meaning
PORT, HOST	3000, 127.0.0.1	Listening address (local only by default)
TWELVEDATA_API_KEY	empty	Free key; empty means labelled sample data
PRICE_SOURCE	binance	binance or mock
NEWS_SOURCE, CALENDAR_SOURCE	rss, faireconomy	Use sample for offline demos
TIMEFRAMES	H1,H4	Timeframes analysed
HISTORY_YEARS	8	Years of candles to load
EVENT_PAUSE, _BEFORE_MIN, _AFTER_MIN, _IMPACT	on, 30, 30, high	Event pause window
LAB_AUTO	on	Daily automatic lab re-runs
NEWS_FEEDS	built-in	Comma-separated feed URLs replacing the defaults
DB_PATH	data/xauusd.db	SQLite file

7.4 Future hosting (not yet built)

Always-on hosting would let alerts and the journal run while the PC is off. Considerations before doing it:

- **Free hosts often sleep idle apps**, which would break the live stream and the journal. Check each host's current limits first.
- **Persistent storage** is needed for the SQLite file (or move to a hosted database).
- **Authentication and HTTPS** are required once the site is reachable from the internet. Today it has none because it is local-only.
- **Secrets** must go in the host's environment settings, never in the repository.

7.5 Release checklist

Step	Check
1	npm test passes (70 tests)
2	Fresh extraction installs and starts; page returns HTTP 200
3	No .env or data folder in the package
4	Version and changelog updated; footer shows the version
5	README links and screenshots display on GitHub; CI is green

8. Phase 6: Maintenance

8.1 Routine

When	Task
Daily (automatic)	Lab re-runs, news and calendar refresh, candles update on their schedules
Weekly	Open the data-source strip at the bottom: all four should be healthy. Review the journal and any alerts
Monthly	Re-check free-tier limits (Twelve Data, Binance, feeds); download journal CSV as a backup of results; copy the data folder
After each update	Run START.bat once, confirm the footer version, check /api/status

8.2 Backups and upgrades

- **Backup:** copy the data folder (xauusd.db) and the .env file somewhere safe. Candles can be re-downloaded; the journal cannot, so back it up.
- **Upgrade:** extract the new version to a new folder, copy the old .env and data folder into it, then run START.bat. Older databases upgrade automatically.
- **Reset:** deleting the data folder starts a fresh database; history reloads from the provider.

8.3 Troubleshooting

Symptom	Likely cause and fix
Yellow banner says sample data	No key saved. Use Settings, the yellow box or SET-KEY.bat
Page looks old or key box missing	An old copy is still running. Close all black windows, run START.bat once; the footer shows the version
Ticker stays on CONNECTING	Binance blocked or offline. Set PRICE_SOURCE=mock to test, or try another network
News card empty	Open /api/status to see which feeds failed; feeds can be blocked or rate-limited
History not loading	Open /api/status: backfill shows paused with a reason; it retries hourly. The free plan allows 8 requests a minute
Lab shows nothing	Needs about 400 cleaned bars; wait for candles or history to load
Port in use error	Another program uses port 3000; change PORT in .env

8.4 Monitoring endpoints

/api/status (each source, last success, errors, history coverage), /api/health (uptime and tick age), and /api/debug/backtest?tf=H4 (candle quality, signal counts and a fresh backtest) give a complete picture without reading code.

9. Risks and limitations

Risk or limitation	Impact	Mitigation
No proven strategy	Signals may lose money	Lab and journal; demo account first; clear wording in the interface

Risk or limitation	Impact	Mitigation
Free-tier changes or limits	Stale or missing data	Pacing, caching, last-good data, status page; re-verify limits
PAXG is a proxy for spot gold	Ticker can differ by a few dollars	Labelled in the interface; signals use real XAU/USD candles
Keyword news scoring	Misjudged headlines	Transparent rules; reason shown; sentiment shrunk when evidence is thin
Local-only, PC must be on	No alerts while off	Hosting is on the roadmap
Backtest optimism	Overstated results	R-multiples, cleaning, walk-forward, forward journal

10. Roadmap

- Free always-on hosting with authentication (and re-checking host limits)
- Calendar surprise tag (actual versus forecast) feeding news impact
- Settings page for strategy parameters and an out-of-sample report for new ideas
- Automatic database backups
- Richer news sources and a measured news-to-price study

Appendix A: API reference

Endpoint	Purpose
GET /api/stream	Server-Sent Events: live ticks and update notices
GET /api/status, /api/health, /api/price	Source status and history coverage, uptime, latest tick
GET /api/strategies	Strategies with lab verdicts and live-journal summaries
GET /api/signal, /api/performance, /api/journal	Per strategy and timeframe: latest signal, backtest, live results
GET /api/lab, /api/lab/summary; POST /api/lab/run	Test lab results and starting runs (local only)
GET /api/news, /api/sentiment, /api/events, /api/pause	News with impact, sentiment, calendar, event-pause state
GET /api/notifications; POST /api/notifications/read	On-site alerts
GET /api/export/journal.csv, /api/export/trades.csv	CSV downloads
GET /api/debug/backtest	Data-quality and signal check
POST /api/settings/twelvedata-key	Validate and save the key (local only)

Appendix B: Glossary

Term	Meaning
XAU/USD	The price of one ounce of gold in US dollars
Candle, H1, H4	A bar showing open, high, low and close for a period of 1 or 4 hours
Stop loss, take profit	Prices at which a trade is closed at a loss or at a target
R (R-multiple)	A trade's result in units of the amount risked: +1R won as much as risked, -1R lost the stop
Profit factor	Total winnings divided by total losses; above 1 means profit
Drawdown	The biggest fall from a previous equity peak
Backtest	Replaying a strategy on past data
Look-ahead bias	Accidentally using future information in a backtest, which flatters results
Walk-forward	Tune on the past, test on the next unseen block, repeat
Bootstrap	Resampling the trades many times to see how much of the result could be luck
Forward test (journal)	Recording real signals as they happen and following them to their outcome
ATR, RSI, EMA	Volatility, momentum and trend-average indicators used by the strategies
Lot	A trade size unit; a standard gold lot is usually 100 ounces

Appendix C: Disclaimer, licence and author

> **Disclaimer.** XAUUSD Pulse provides analysis for personal and educational use. Signals, sentiment and test results are rule-based outputs, not recommendations. Past performance does not guarantee future results, and trading gold carries a risk of loss. Try any idea on a demo account first. This is not financial advice.

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